

Def.

Let G be a group and A be a non-empty subset. A map

$$\alpha: G \times A \rightarrow A: (g, a) \mapsto g \cdot a$$

is called a group action of G on A if it satisfies

- 1) For all $a \in A$, $1_G \cdot a = a$
- 2) For all $g_1, g_2 \in G$, and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Prop. Given a group action α of G on A , there exists a homomorphism

$$\mathcal{Q}_\alpha: G \rightarrow S_A: g \mapsto \sigma_g \quad \text{where} \quad \sigma_g: A \rightarrow A: a \mapsto g \cdot a.$$

which is called the permutation representation associated to the given action.

Conversely, given a homomorphism $\mathcal{Q}: G \rightarrow S_A$, there exists a group action

$$\alpha_{\mathcal{Q}}: G \times A \rightarrow A: (g, a) \mapsto \mathcal{Q}(g)(a)$$

Proof. We only need to check that the given map is well-defined,

Let $g \in G$ be fixed. Then, $\sigma_g \circ \sigma_{g^{-1}}(a) = \sigma_{g^{-1}} \circ \sigma_g(a) = a$, therefore $\sigma_g \in S_A$.

The converse is obtained by the definition of group action.

Note: The concept of group action is: for a set A , determine a permutation

$$A \rightarrow A: a \mapsto g \cdot a$$

from each element $g \in G$, compatible with the group structure of G , in the sense: the afforded homomorphism $\mathcal{Q}_\alpha: G \rightarrow S_A$ preserves the operation.

Def. Let G acts on A . For each $a \in A$, a subset

$$\text{Orb}_G(a) \stackrel{\text{def}}{=} \{g \cdot a \mid g \in G\} \subseteq A$$

is called the Orbit of a .

Prop. Let a group G acts on a set A . Given $a \in A$, a map

$$G/G_a \rightarrow \text{Orb}_G(a): gG_a \mapsto g \cdot a$$

is bijection. In particular, if G is finite, then

$$|G/G_a| = \#\text{Orb}_G(a).$$

Proof. Given $g_1, g_2 \in G$,

$$g_1 G_a = g_2 G_a \Leftrightarrow g_1^{-1} g_2 \in G_a \Leftrightarrow g_1^{-1} g_2 a = a \Leftrightarrow g_1 a = g_2 a$$

So it is well-defined and injective. And the surjectiveness is clear. ~~is~~

Thm.

Let G be a group and $H \leq G$ be a subgroup.

Consider G acts on $\{gH \mid g \in G\}$ by left multiplication, and affords homomorphism

$$\pi_H: G \rightarrow S_{G/H}: g \mapsto \sigma_g \text{ where } \sigma_g(xH) = gxH.$$

Then G acts transitively on A and the stabilizer $G_{1 \cdot H} = H$, and

$$\ker \pi_H = \bigcap_{x \in G} xHx^{-1} \leq H$$

is the largest normal subgroup of G contained in H .

Proof. Given $aH, bH \in A$, $bH = ba^{-1}(aH)$, so aH and bH lie in same orbit.

And, since $gH = H \iff g \in H$,

$$G_{1 \cdot H} = \{g \in G \mid gH = H\} = H.$$

To show the last assertion, by definition,

$$\ker \pi_H = \bigcap_{x \in G} G_{xH}$$

and $G_{xH} = \{g \in G \mid gxH = xH\} = \{g \in G \mid x^{-1}gx \in H\} = xHx^{-1}$.

Moreover, $\ker \pi_H \leq H$ because it is kernel of the homomorphism, and for any normal subgroup $N \leq G$ of G contained in H ,

$$\text{for all } x \in G, N = xNx^{-1} \leq xHx^{-1}$$

therefore $N \leq \ker \pi_H$. $\square$

Prop. If $H \leq G$ has finite index $n = |G:H|$, then there is a normal subgroup $K \leq G$ s.t.

$$K \leq H \text{ and } |G:K| \mid n!$$

Proof. The kernel of $\pi_H: G \rightarrow S_{G/H}: g \mapsto (xH \mapsto gxH)$ is contained in H . And $G/\ker \pi_H \cong \text{Im } \pi_H \leq S_{G/H} \cong S_n$. Thus $|G:\ker \pi_H| \mid n!$, and

$$|G:\ker \pi_H| = |G:H| \cdot |H:\ker \pi_H| = n \cdot |H:\ker \pi_H|$$

so $|H:\ker \pi_H|$ divides $(n-1)!$.

Thm.

Let G be a finite group of order n , and let p be the smallest prime dividing n . Then, every subgroup of index p is normal.

Proof. Let $H \leq G$ with $|G:H| = p$. Let G acts on left cosets of H and

$$\pi_H: G \rightarrow S_{G/H}$$

be the afforded permutation of the given action.

Then, $\ker \pi_H \leq H$, let $K = \ker \pi_H$ and $p = |H:K|$. Then,

$$|G:K| = |G:H| \cdot |H:K| = p^2.$$

Since $G/K \cong \text{Im } \pi_H \leq S_p$, so p^2 divides $p!$ by the Lagrange's Theorem.

Therefore, p divides $(p-1)!$. But all prime factors of $(p-1)!$ are less than p , by the minimality of p , p must be 1, that is, $H = K = \ker \pi_H \trianglelefteq G$.

Note: let G acts on finite set A which has m elements.

Then, the afforded permutation $\varphi: G \rightarrow S_m$ satisfies:

$$G/\ker \varphi \cong \text{Im } \varphi \leq S_m \text{ implies } |\text{Im } \varphi| \mid |G| \text{ and } |\text{Im } \varphi| \mid m!$$

So, if p is the smallest prime factor of $|G|$, then

$$m < p \Rightarrow \text{Im } \varphi = 1 \quad \text{and} \quad m = p \Rightarrow |\text{Im } \varphi| \in \{1, p\}.$$

In the case of the above theorem, the action is transitive, therefore $\text{Im } \pi_H \neq 1$. Therefore, $|\ker \pi_H| = |H|$ and $\ker \pi_H \leq H$, so proved.

Def. Let G be a group, and let $H \leq G$ be a subgroup.

A subgroup $\text{Core}_G(H) \stackrel{\text{def}}{=} \bigcap_{g \in G} gHg^{-1}$ is called the core of H in G . (It is normal).

Note: the theorem above gives

$$H \trianglelefteq G \Leftrightarrow H = \text{Core}_G(H).$$

Problems in dummit, 4.2.

Thm. If G is a group of order p^n where p is prime and $n \in \mathbb{N}$, then every subgroup of index p is normal.

Proof. Let $H \leq G$ be the subgroup with $|G:H| = p$.

Then, the Homomorphism defined by the left Coset action

$$\varphi: G \rightarrow S_{G/H}; g \mapsto (xH \mapsto gxH)$$

satisfies

$$G/\ker \varphi \cong \text{Im } \varphi \leq S_{G/H}$$

So, $|G/\ker \varphi|$ divides $p!$, but since p is prime, $|G/\ker \varphi| \in \{1, p\}$.

Since $\ker \varphi \leq H \neq G$, $\ker \varphi \neq G$, so $|G/\ker \varphi| = p$, that is, $|\ker \varphi| = p^{n-1}$.

Consequently, $H = \ker \varphi \trianglelefteq G$ is normal. $\square$

Notes: If G is group of order p^2 , then it has order p element $x \in G$ by Cauchy's theorem, and $\langle x \rangle \trianglelefteq G$.

Thm. Group of order 6 are only $\mathbb{Z}/6\mathbb{Z}$ and S_3 .

Proof. If G is abelian, then the Cauchy theorem gives order 2 element x and order 3 element y . Since x and y commute, $|xy| = 6$, so $G = \langle xy \rangle \cong C_6$.

Assume that G is non-abelian. By the Cauchy theorem again, there exists order 2 element x . Let $H = \langle x \rangle$. Then, the homomorphism

$$\tau_H: G \rightarrow S_{G/H}; g \mapsto (xH \mapsto gxH)$$

satisfies that $G/\ker \tau_H \cong \text{Im } \tau_H \leq S_{G/H} \cong S_3$.

If H is normal, then for all $g \in G$, $gxg^{-1} \in \langle x \rangle$, that is, $gxg^{-1} = x$, so x commutes with order 3 element y , it implies $\langle xy \rangle = 6$, it contradicts. Therefore H is non-normal, so $\ker \tau_H \neq H$, the trivial subgroup. Now,

$$G/\ker \tau_H \cong G \cong \text{Im } \tau_H \leq S_{G/H} \cong S_3$$

and since G and S_3 both are order 6, thus $G \cong S_3$ $\square$

Thm. Let G be a finite group and let $\pi: G \rightarrow S_G$ be the left regular representation. Given $x \in G$, the permutation $\pi(x)$ is a product of $|G|/|x|$ numbers $|x|$ -cycles.